

Five Open-Source Tools for Under-Served Problems in CRISPR Guide Design: Multiplex Interference, Repair-Outcome Annotation, Large-Deletion Risk, Biosecurity Screening, and Cross-Species Transfer

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Abstract

Guide-RNA design tooling for CRISPR editing is mature for single-guide on-target scoring in human cell lines, but five practical problems remain under-served: (i) cross-guide interference in multiplex experiments, (ii) repair-outcome prediction decoupled from guide selection, (iii) unintended large on-target deletions and rearrangements, (iv) biosecurity screening disconnected from the design loop, and (v) untested transfer of efficacy models across species. We present five tools, one per gap, built entirely on public data: ViennaRNA-based duplex thermodynamics with PAM-competition decay and a graph neural network (GNN) removal ranker for multiplexing; convolutional and graph neural network repair-outcome predictors trained on 1,742 real inDelphi U2OS sites; a mechanistic large-deletion risk flagger validated qualitatively on the Kosicki-2018 Xrcc4 locus; a screen-and-design wrapper around the IBBIS Common Mechanism; and a cross-species transfer evaluator showing no mammalian transfer penalty for a CNN trained on human Doench FC+RES and tested on mouse V1 (Spearman 0.78 vs 0.63). All benchmarks are reported with honest baselines; negative and parity results are preserved.

1 Introduction

2 Data

2.1 inDelphi U2OS LibA

Event-level counts from [?] (figshare 6837956, U2OS_+ LibA_postCas9_rep1.csv) joined to 57-nt target contexts (targets-libA.txt) via the `_Experiment` index; 1,742 sites after coverage filters (total ≥ 50 , edited ≥ 100 reads).

2.2 Doench 2016 FC+RES and V1

Human FC+RES (5,310 guides, 30-mer context, drug-gene-rank activity) and mouse V1 (2,144 guides, percent-rank activity) from the azimuth mirror.

2.3 crisprSQL

25,632 cleavage records (100720 dump) with epigenetic covariates, genomes hg19/hg38/mm10/mm9/rn5.

2.4 Real-locus sequence

Mouse Xrcc4 mRNA (NM_028012.4) fetched live from NCBI for risk-flagger validation.

3 Tool 6: Multiplex cross-guide interference screener

3.1 Model

For guides g_i, g_j , the duplex channel uses ViennaRNA lorenz2011 heteroduplex MFE $\Delta G_{ij}^{\text{dup}}$ (duplexfold) and the longest contiguous antiparallel complementarity run c_{ij} .

Definition 1 (PAM competition). *Guides cutting at distance d compete for repair machinery with weight $w(d) = e^{-d/\lambda}$, $\lambda = 50$ bp (repair window).*

Definition 2 (Interference graph). *Weighted graph on guides with edge weight $a_{ij} = 0.6 \min(1, |\Delta G_{ij}^{\text{dup}}|/20) + 0.4 w(d_{ij})$. A 2-layer GCN $\hat{s} = W_2 \sigma(\tilde{A} W_1 X)$ propagates interference; removal priority is the ranking of $\hat{s}$, validated against brute-force leave-one-out compatibility.*

3.2 Results

CNN/GNN surrogates for single-strand fold MFE reach Spearman 0.63/0.64 against the ViennaRNA teacher (ridge on 3-mers 0.65): parity, not improvement; ViennaRNA remains the primary engine (Table ??).

Table 1: Duplex-MFE surrogate benchmark (2,000 spacers, 400 held out).

Model	Spearman	Pearson
Ridge (3-mer)	0.648	0.653
CNN (one-hot)	0.633	0.658
GCN (3-mer graph)	0.645	0.645

4 Tool 7: Repair-outcome annotator

4.1 Model and math

Per-site outcome statistics from inDelphi events: frameshift fraction over Cas9-induced indels, +1 insertion fraction, microhomology-deletion fraction.

Proposition 1 (Microhomology score). *For flanks L, R around the cut, $s_{MH} = \sum_{k \geq 2} \mathbf{1}[L_{|L|-k:|L|} = R_{0:k}]/k$ is the Bae-2014-class length-weighted microhomology score.*

4.2 Results

Held-out Spearman (349 test sites): +1-insertion 0.60 (GBT positional) / 0.50 (CNN); frameshift 0.32 (GBT); microhomology-deletion 0.23 (GBT). Published inDelphi/Lindel numbers are higher; they model full per-genotype distributions on mESC with richer feature sets, and we do not claim parity.

5 Tool 8: Large-deletion risk flagger

Mechanistic, per [?] and [?]: direct-repeat pairs spanning the cut (worst-case deletion span), sequence complexity, homopolymers. On the real Xrcc4 locus (NM_028012.4): 18/43 scanned cut sites flagged MODERATE, predicted spans to 500 bp - qualitatively consistent with the kilobase-scale deletions measured at this locus. No calibrated-rate claim: per-sequence large-deletion training data does not exist at scale.

6 Tool 9: Biosecurity screen-and-design wrapper

Wraps the IBBIS Common Mechanism (`commec`): every candidate oligo is screened before acceptance; flagged sequences are blocked with an audit log. When `commec` is unavailable the wrapper raises rather than silently passing unscreened sequences. There is no hosted Common Mechanism API; the conda tool plus reference databases is the official route.

7 Tool 10: Cross-species transfer evaluator

Table 2: Human (FC+RES) $\rightarrow$ mouse (V1) transfer, Spearman.

Model	Human held-out	Mouse transfer	Drop
Ridge (3-mer)	0.279	0.331	-0.052
CNN (one-hot 30-mer)	0.630	0.781	-0.151
GCN (3-mer graph)	0.289	0.281	+0.008

The CNN transfers from human to mouse *without penalty* (Table ??); the apparent gain is a label-noise asymmetry between the FC+RES drug-gene-rank labels and V1 percent-rank labels,

not improved biology. Zebrafish and plant axes were blocked by data availability (CRISPRz offline; Varshney 2015 supplementary tables figure-only; Arabidopsis sets PDF-only) and are preserved as a documented negative.

8 Limitations

Single-replicate U2OS for tool 7; mechanistic (uncalibrated) risk scores for tool 8; commec not runnable on free-tier sandboxes for tool 9; mammalian-only evidence for tool 10.

9 Conclusion

References

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